

KNOWLEDGE 15	INQUIRY 20	COMMUNICATION 11	APPLICATION 9
-----------------	---------------	---------------------	------------------

PART B: SHORT ANSWER QUESTIONS

Answer the following questions in the space provided.

Thinking/ Inquiry (20 marks total)

1. For each of the following molecules, do the following: (12 marks)

- a) Draw the Lewis structure indicate the geometric shape as predicted by the VSEPR Theory
 b) Determine if the molecule is polar or non-polar
 c) Indicate the hybridization around the **central atom**

Compound	Lewis Structure in VSEPR Shape (incl. bond dipoles and dipole vectors)	Polar/Non-Polar	Hybridization of Central Atom
CO ₃ ²⁻		$\Delta EN = (3.5) - (2.5) = 1.0$ Polar bonds Dipole V. cancel out non-polar	sp ²
AsH ₅		$\Delta EN = (2.1) - (2.0) = 0.1$ Non-polar	sp ³
PF ₃		$\Delta EN = (4.0) - (2.1) = 1.9$ Polar bonds Polar	sp ³

2. a) Cobalt can have a valence of +2 and +3. Write the **electron configuration** for the atom, as well as the electron configurations for each of the possible ion charges above. You may use the shorthand (noble gas) configuration. (3 marks)

Cobalt atom: [Ar] 4s² 3d⁷ ✓Co²⁺: [Ar] 3d⁷ ✓Co³⁺: [Ar] 3d⁶ ✓b) List the four quantum numbers for the last electron of the **cobalt atom**. (1 mark)

n = 3 ✓

m_s = -1/2 ✓

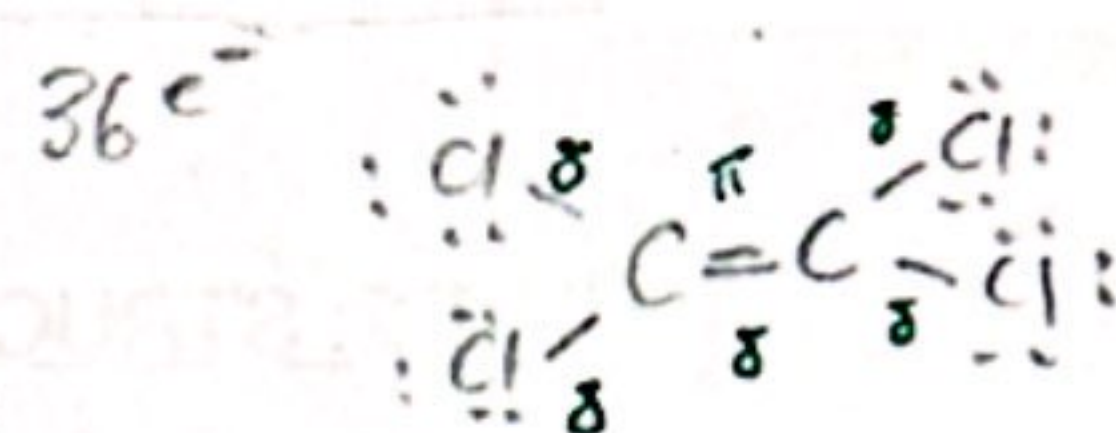
l = 2 (d orbital)

3, 2, -1, -1/2

m_l = -2 → 2 = -1 ✓

1 0 1 1 1

3. For a molecule of C_2Cl_4 complete the following: (4 marks total)



(a) Draw the Lewis structure in the space to the right (1 mark)

(b) What is the hybridization of the central atom? sp^2 (1 mark)

(c) What is the VSEPR shape name at each central atom. trigonal planar (1 mark)

(d) Identify any sigma or pi bonds by labelling them on your Lewis structure that you drew in part (a). (1 mark)

COMMUNICATION (11 marks total)

1. Use the concepts of Lewis Structure in VSEPR shape and intermolecular forces, determine which of these 2 molecular compounds have the higher boiling point - SiO_2 or SF_2 . Complete the chart below. (5 marks)

Compound	Lewis Structure in VSEPR shape	Intermolecular Forces
SiO_2 16 e ⁻	 non-polar	London dispersion (LDF)
SF_2 20 e ⁻	 polar	London dispersion forces dipole-dipole forces
Which has the higher boiling point and explain	SF_2 has a higher boiling point because it has stronger dipole-dipole forces meaning its intermolecular forces are stronger and it will be harder to get it into the gaseous state as the molecules will better stick together. ∴ Higher energy to remove = higher boiling point	

2. What is wrong with each of the following quantum number sets for elements on the periodic table? Briefly explain the error. (2 marks)

a) 2, 0, +1, -1/2

m_l can only be $-l \rightarrow l$
since $l=0$ $m_l \neq +1$

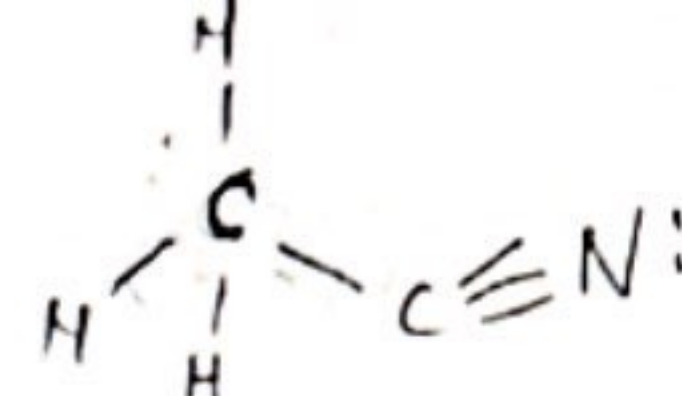
b) 7, 3, -2, +1/2

energy level 7 only exists for the s orbital
the l value indicates the electron is in an f orbital
but the highest energy l f orbital is 5

3. Match the solids, $Al_{(s)}$, $MgSO_{4(s)}$, $CO_{2(s)}$, and $SiC_{(s)}$ to the property listed below. (4 marks)

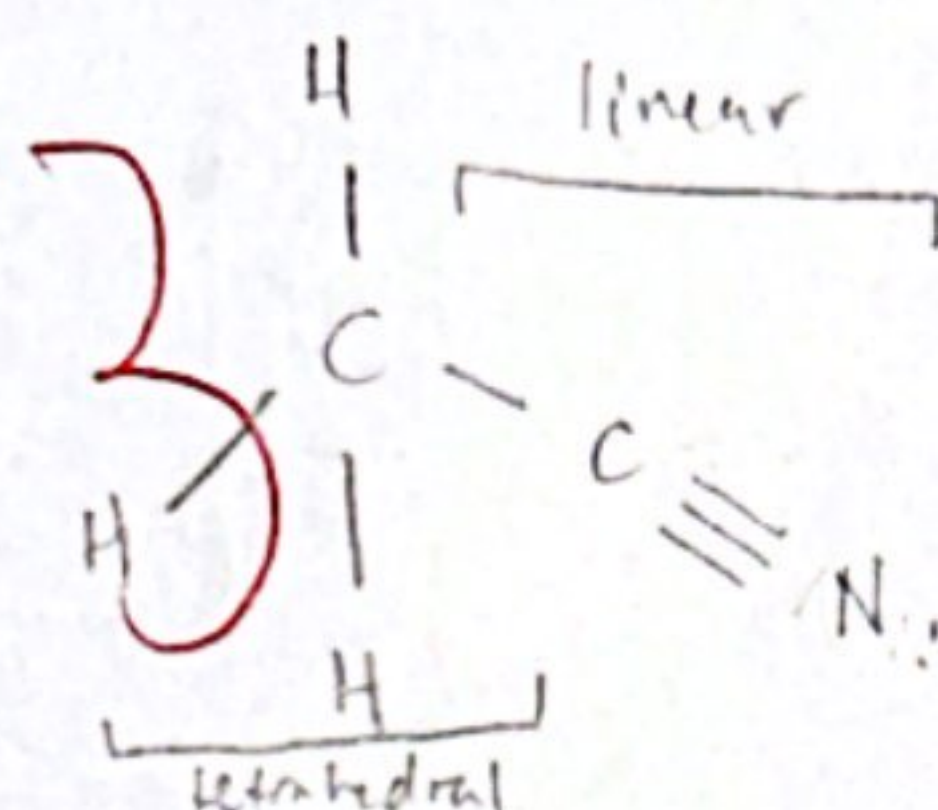
Compound/ Element	Structure (Crystal Type/Name) names	Property
a) $Al_{(s)}$	metallic compound	High melting point, conducts electricity
b) $CO_{2(s)}$	molecular compound	Low melting point, soft
c) $SiC_{(s)}$	network covalent compounds	Very high melting point, nonconductor
d) $MgSO_{4(s)}$	ionic compound	High melting point, soluble in water

APPLICATION (9 marks total)



VSEPR

1. a) Draw the Lewis structure for methyl cyanide (CH_3CN). Predict the shapes individually around the carbon atoms (3 marks)

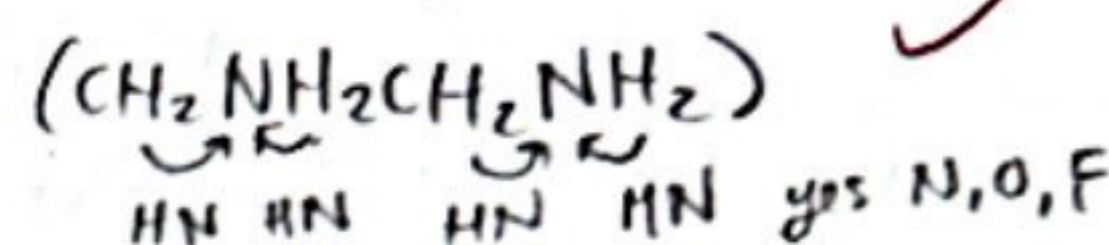
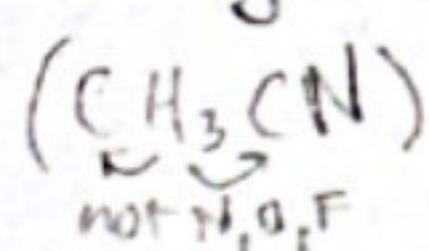


tetrahedral
4 bonded pairs
0 lone pairs

linear
2 bonded pairs
0 lone pairs

b) methyl cyanide CH_3CN and diaminoethane ($\text{CH}_2\text{NH}_2\text{CH}_2\text{NH}_2$) have similar molecular masses. Explain why the boiling point of methyl cyanide is 82°C and diaminoethane is 116°C . (1 mark)

methyl cyanide does not have hydrogen bonding while diaminoethane does



Since hydrogen bonds are the strongest intermolecular force, it means diaminoethane can form stronger bonds with itself than methyl cyanide. Since the boiling point refers to the temperature at which there is enough energy to detach a substance into a gas, it will take more energy to break the intermolecular forces on diaminoethane and therefore it has a higher boiling point.

Investigation 2A Questions

2. Account for the attraction of the liquid streams to the charged rod:

a) What does the attraction of the liquid streams to the charged rod suggest about the molecules? (1 mark)

That the molecules are polar. (+) end $\rightarrow$ (-) charged rod

b) Why do the different liquids experience different amount of deflection (1 mark)

It depends on how polar a molecule is, the "more positive" a molecule's end gets, the more it is attracted to the rod and the further it will bend to reach it

3. Rank these substances in order of highest viscosity to lowest viscosity and explain why:

propan-1-ol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$), propane-1,2-diol ($\text{CH}_3\text{CH}_2\text{OHCH}_2\text{OH}$), propane-1,2,3-triol ($\text{CH}_2\text{OHCHOHCH}_2\text{OH}$) (3 marks)

Highest: propan-1,2,3-triol Middle: propane-1,2-diol lowest: propan-1-ol

The more stronger bonds something can make, the more viscous (higher attraction to itself meaning it won't want to flow elsewhere as easily)

(lowest) propan-1-ol has the ^{least} hydrogen bonding sites

(highest) propane-1,2,3-triol has the most

larger/longer propane-1,2-diol is in the middle

The ^{1st} the molecule, the more likely it is to get tangled on itself and have higher strength London dispersion forces

of H-b. They are all polar

(highest) propane-1,2,3-triol has the longest molecular structure

(lowest) propan-1-ol has the shortest molecular structure

propane-1,2-diol is in the middle